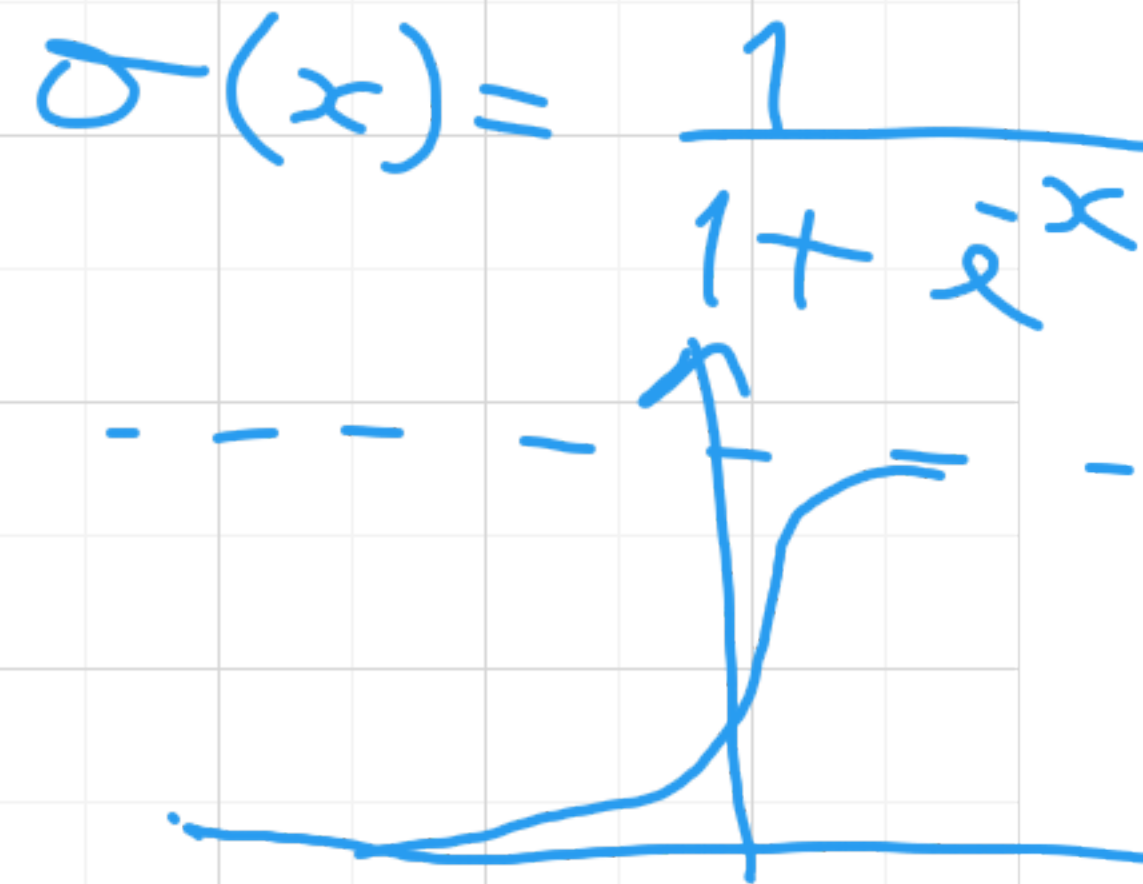


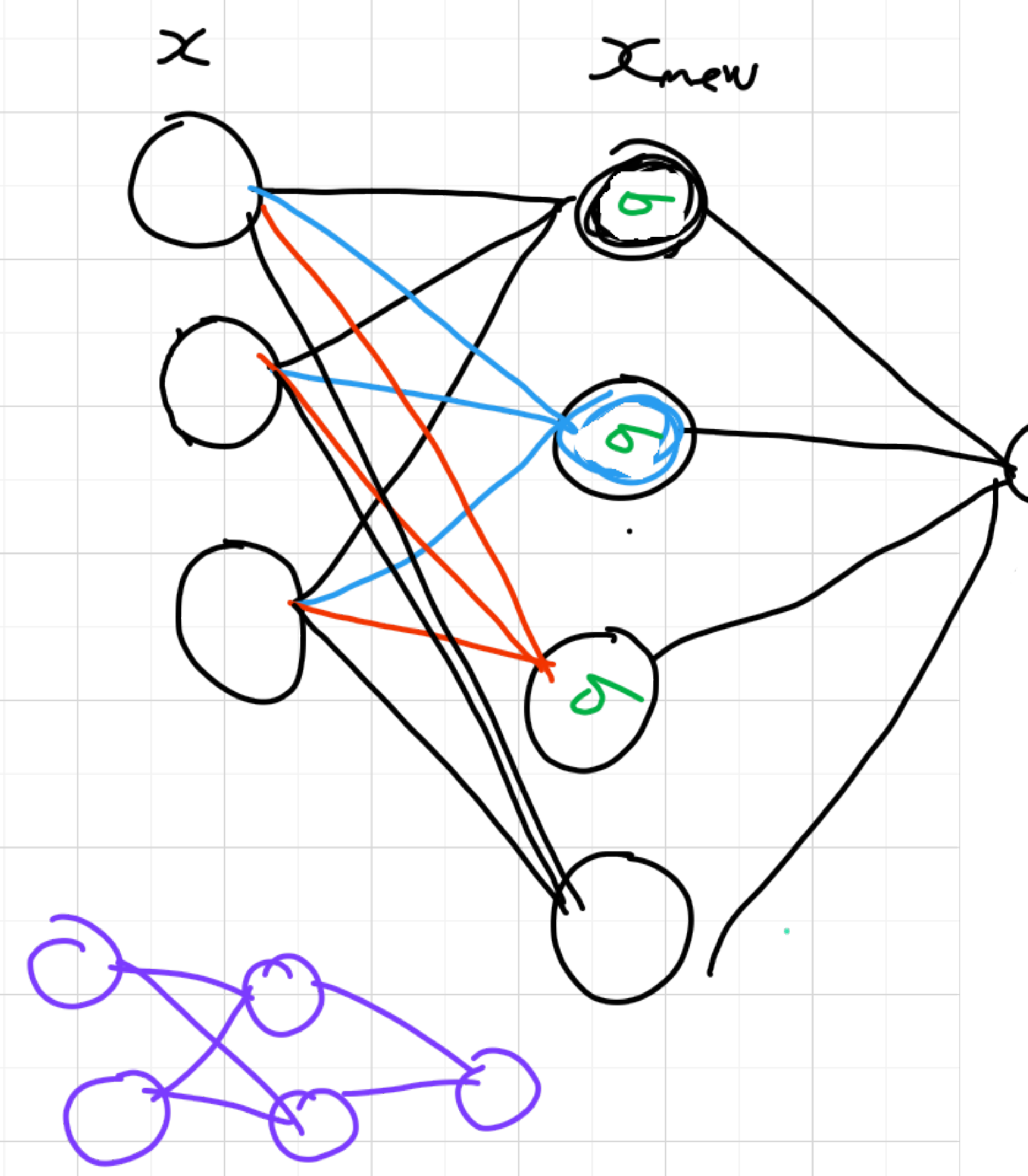
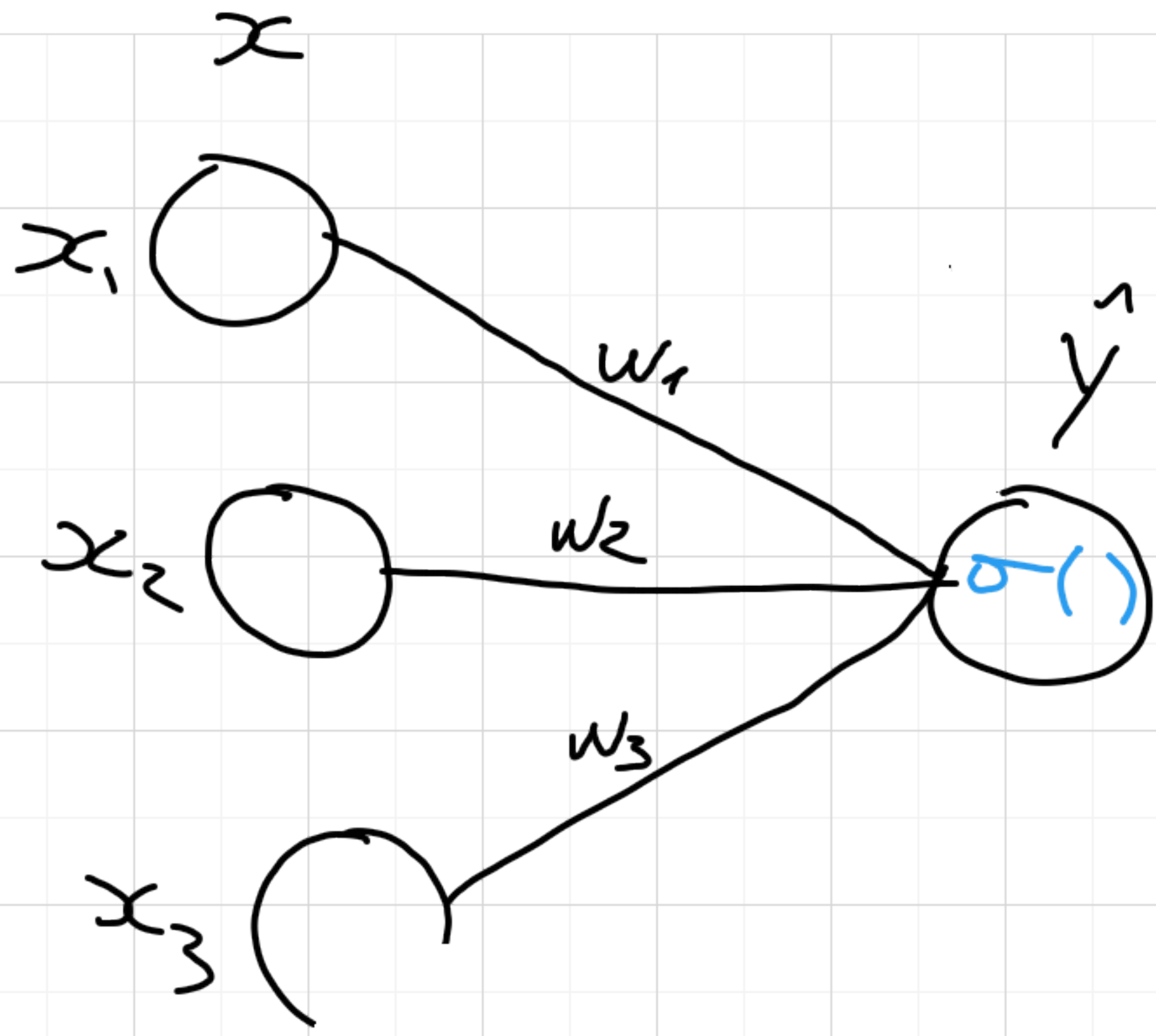
Linear Regression

and Logistic as MLP

Multi-layer Perceptron

$$y = \sigma(w_1 \cdot x_1 + w_2 \cdot x_2 + b)$$





$$x_{new_1} = W_{1 \times 3} \cdot x_{3 \times 1} + b = w_1 \cdot x_1 + w_2 \cdot x_2 + w_3 \cdot x_3 + b$$

$$x_{new_2} = W_{1 \times 3} \cdot x_{3 \times 1} + b$$

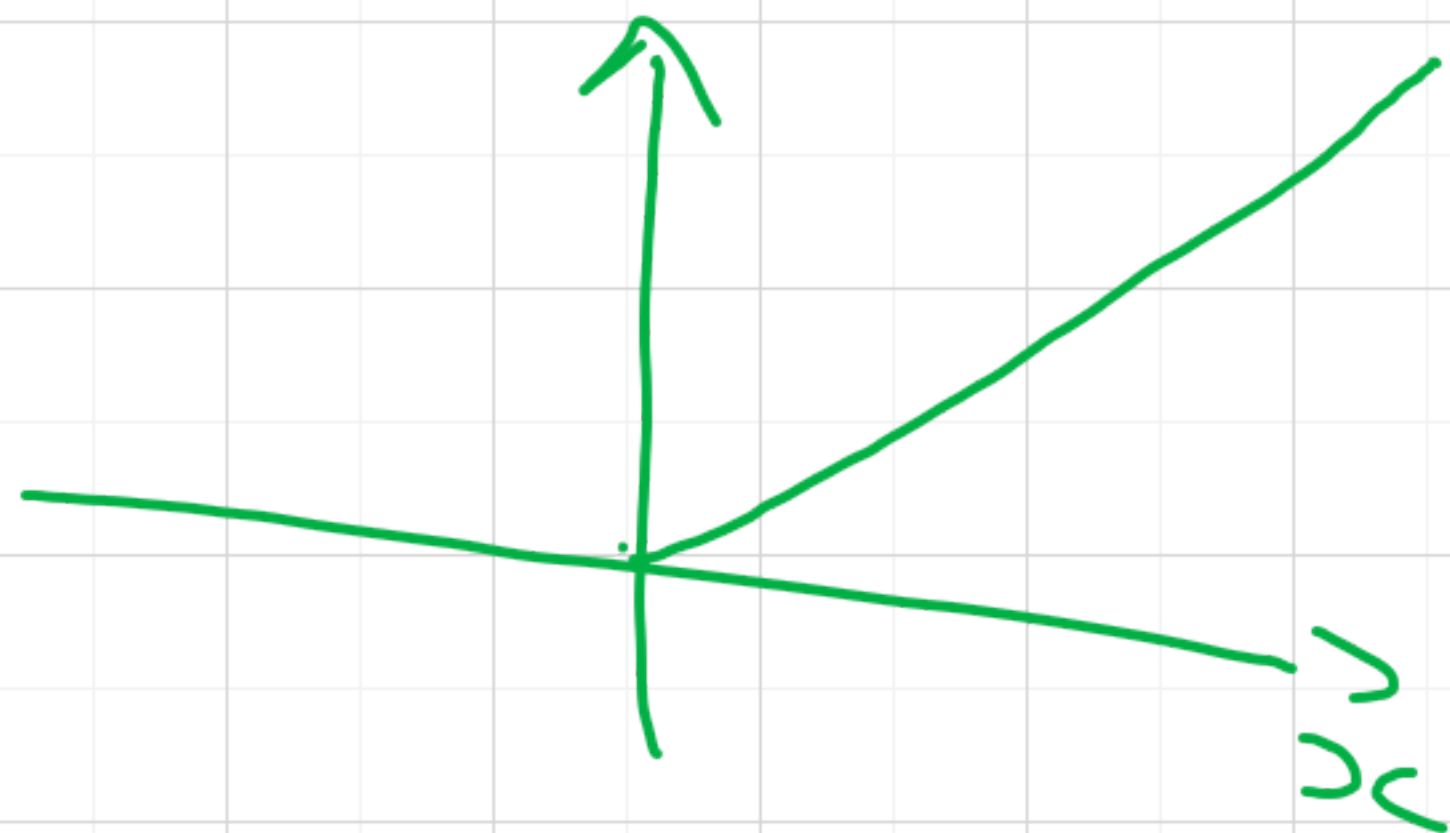
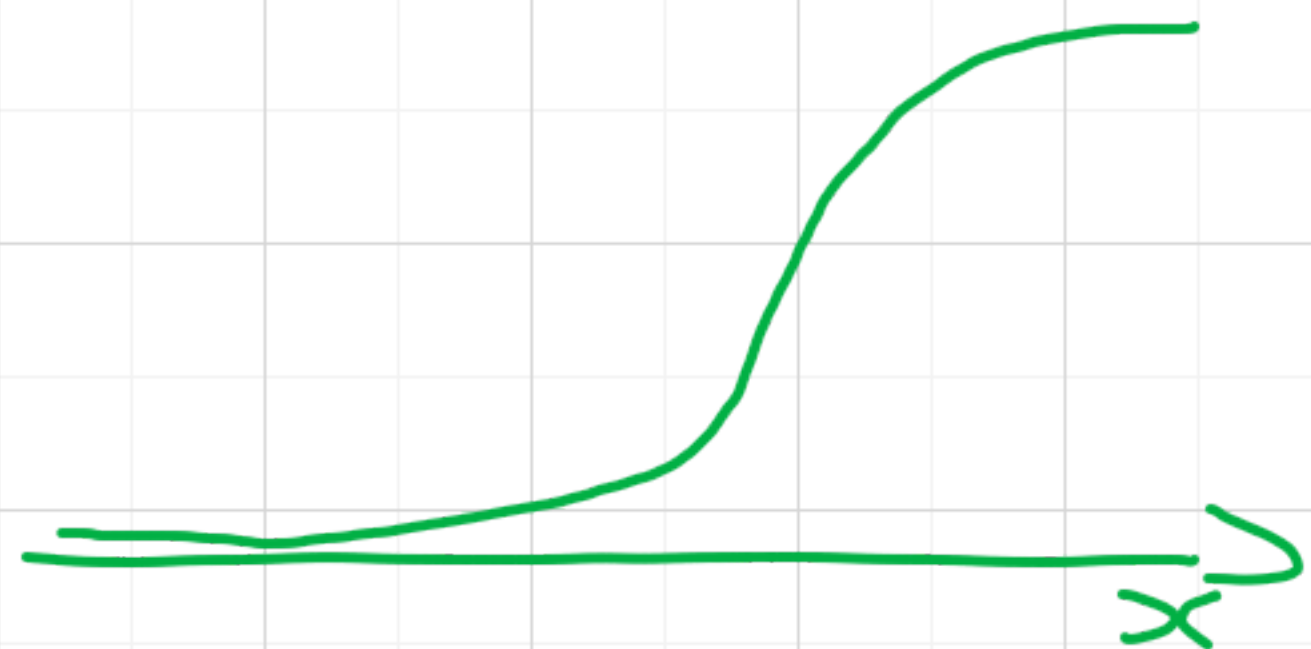
$$x_{new_3} = W_{1 \times 3} \cdot x_{3 \times 1} + b$$

$$\sigma = \begin{cases} \text{Sigmoid} \rightarrow \\ \tanh(x) \\ \text{ReLU}(x) \\ \vdots \end{cases}$$

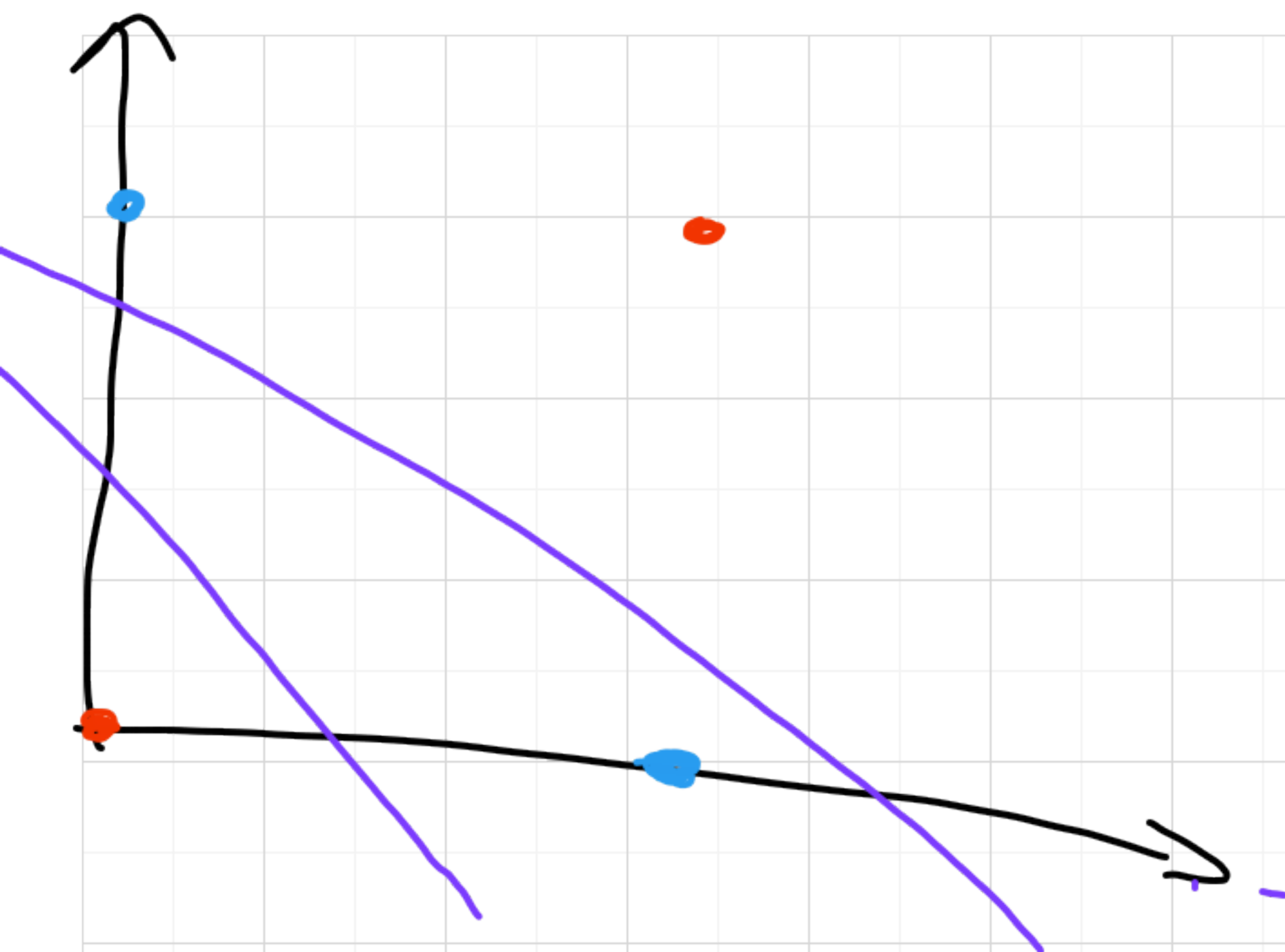
$$\hat{y} = W_{\hat{y}} \cdot x_{new} + b_{\hat{y}} =$$

$$= W_{\hat{y}} \cdot (W_{x_{new}} \cdot x + b)$$

$$= \underbrace{W_{\hat{y}} \cdot W_{x_{new}}}_{W_{new}} x + \underbrace{W_{\hat{y}} \cdot b}_{b_{new}} = W_{new} \cdot x + b_{new}$$



XOR	0	1
0	0	1
1	1	0



||

x
○
○
⋮
○
○

x'
○
○
○
○
○
○

x''
○
○
○

x'''
○

Universal Approximation theorem

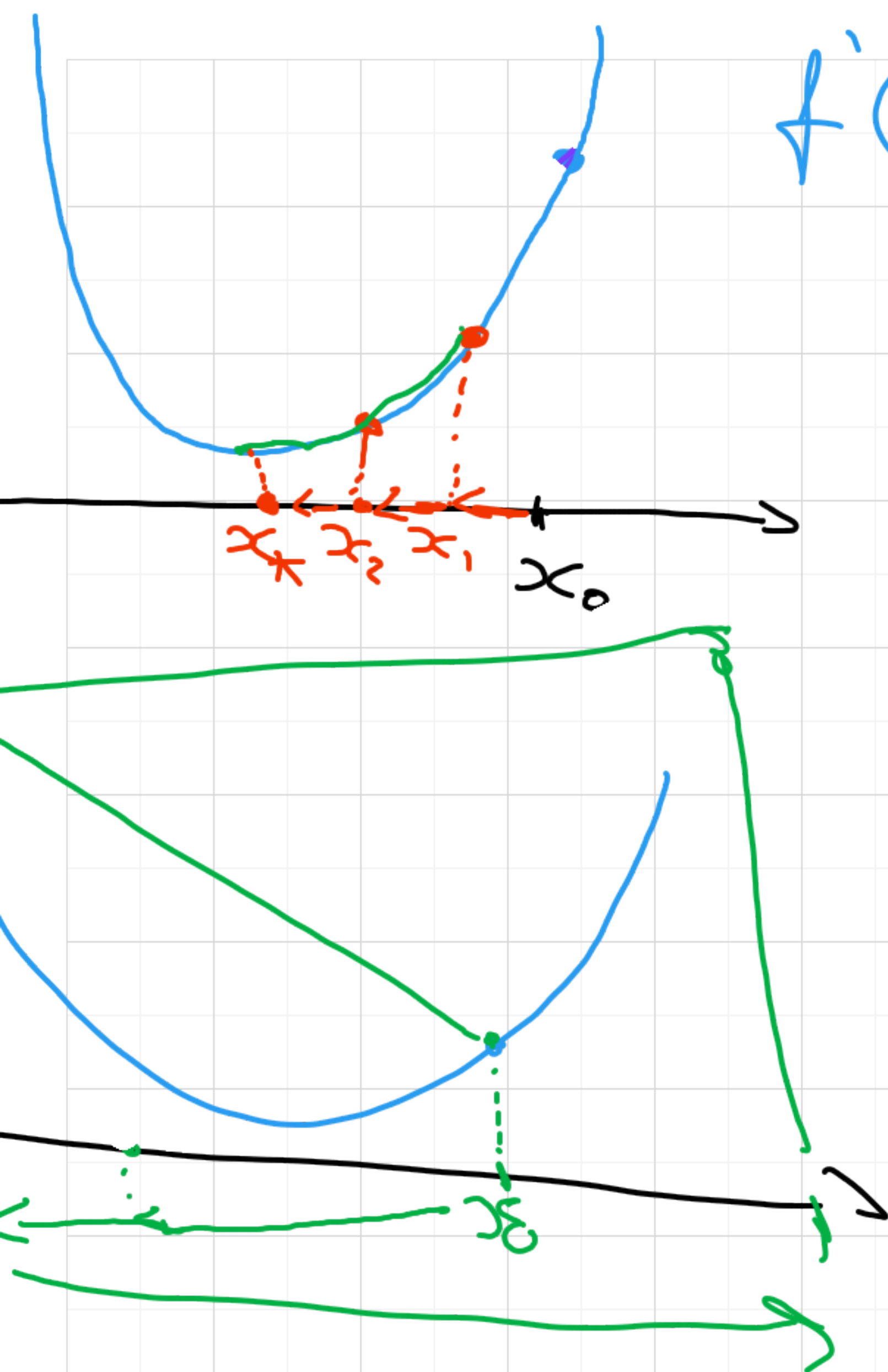
$$f(x) = ?$$

x
○
⋮
○

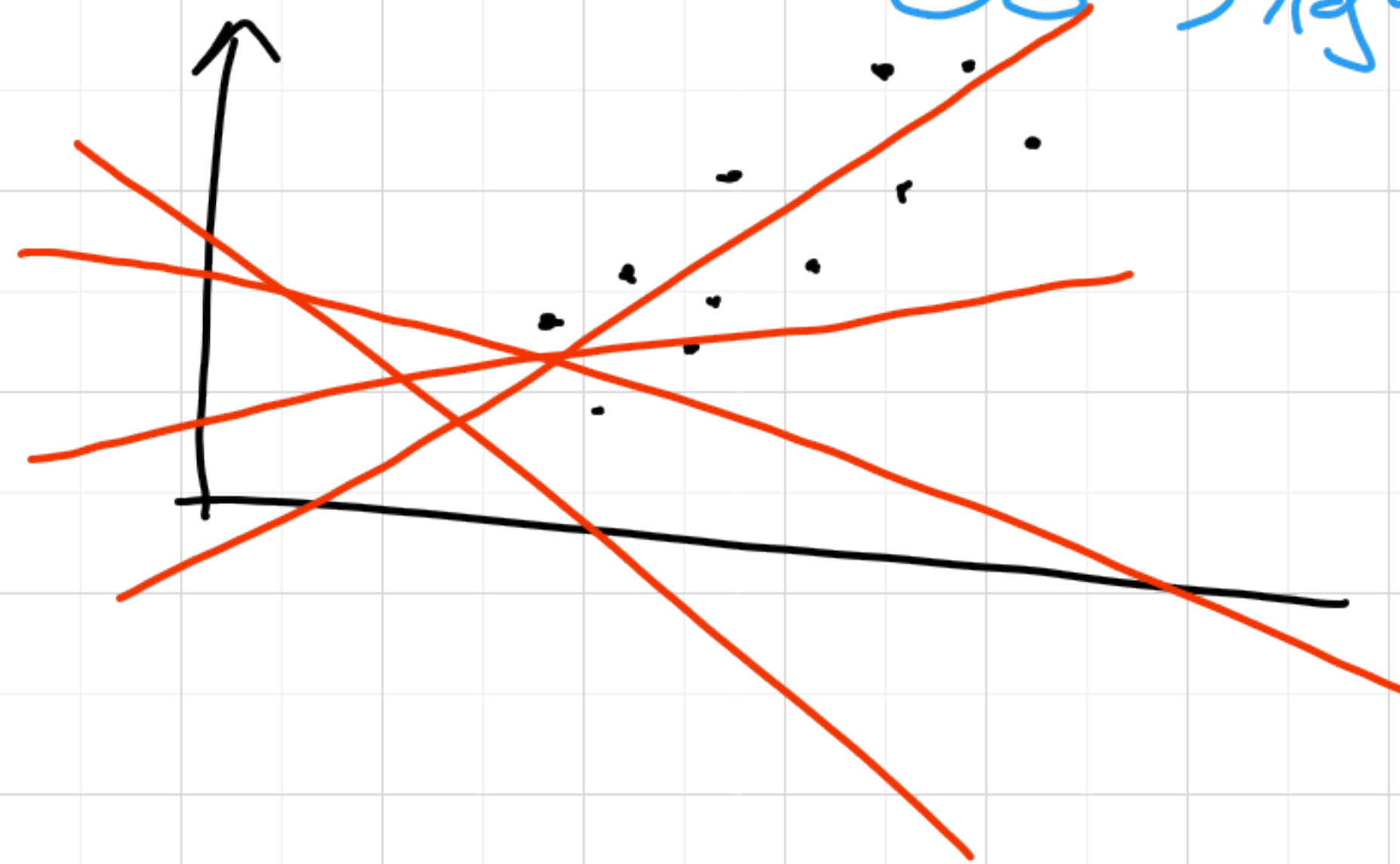
x_{new}
○
⋮
○

y_1





$f'(x_0) > 0 \rightarrow$ Go left
 $< 0 \rightarrow$ Go right

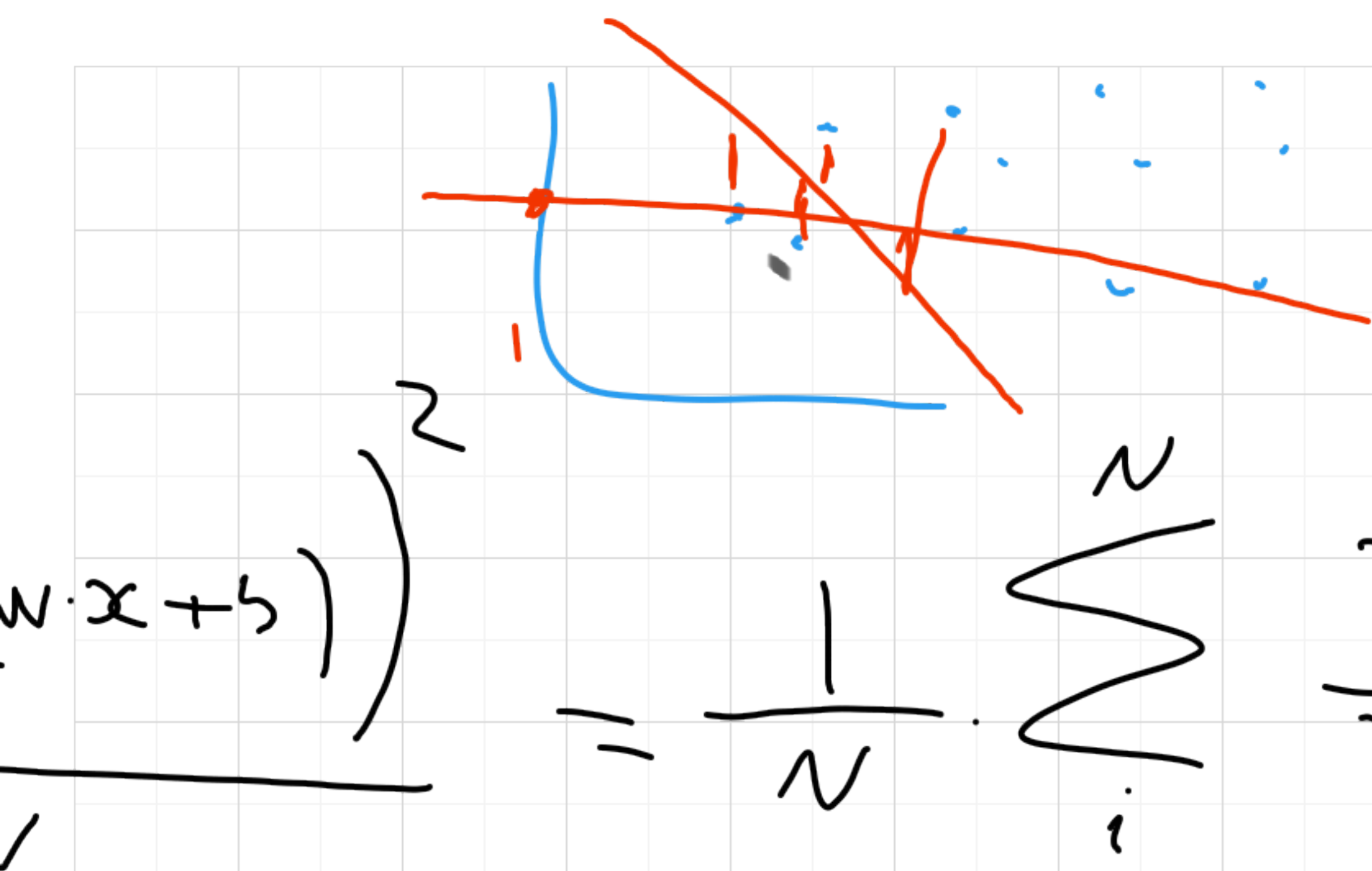


$MSE(y, \hat{y})$

$$\hat{y}_i = w \cdot x_i + b$$

$$J = \frac{1}{N} \sum_i^N (y_i - \hat{y}_i)^2$$

$$\frac{\partial MSE}{\partial w} = \frac{\partial}{\partial w} \sum_i^N (y_i - \hat{y}_i)$$



$$\frac{\partial}{\partial x} \mu(x) = m \cdot \partial \mu \cdot \mu^{m-1}$$

M

$$= \frac{1}{N} \sum_i^N$$

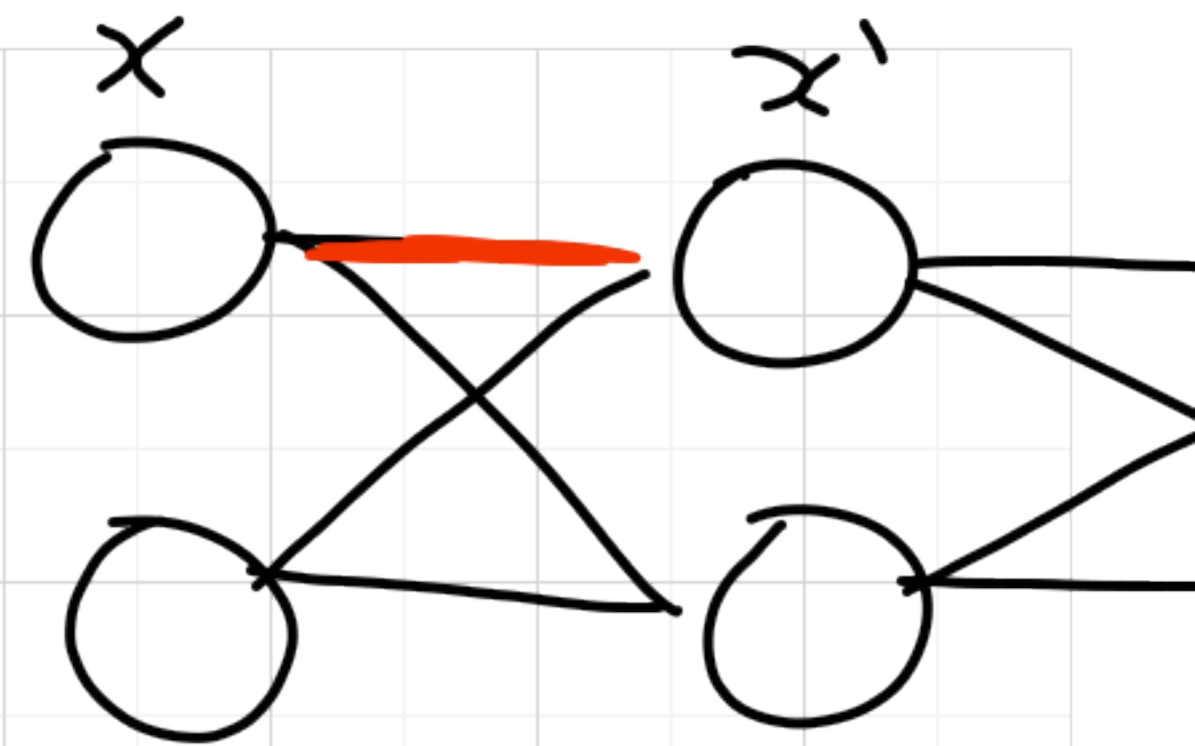
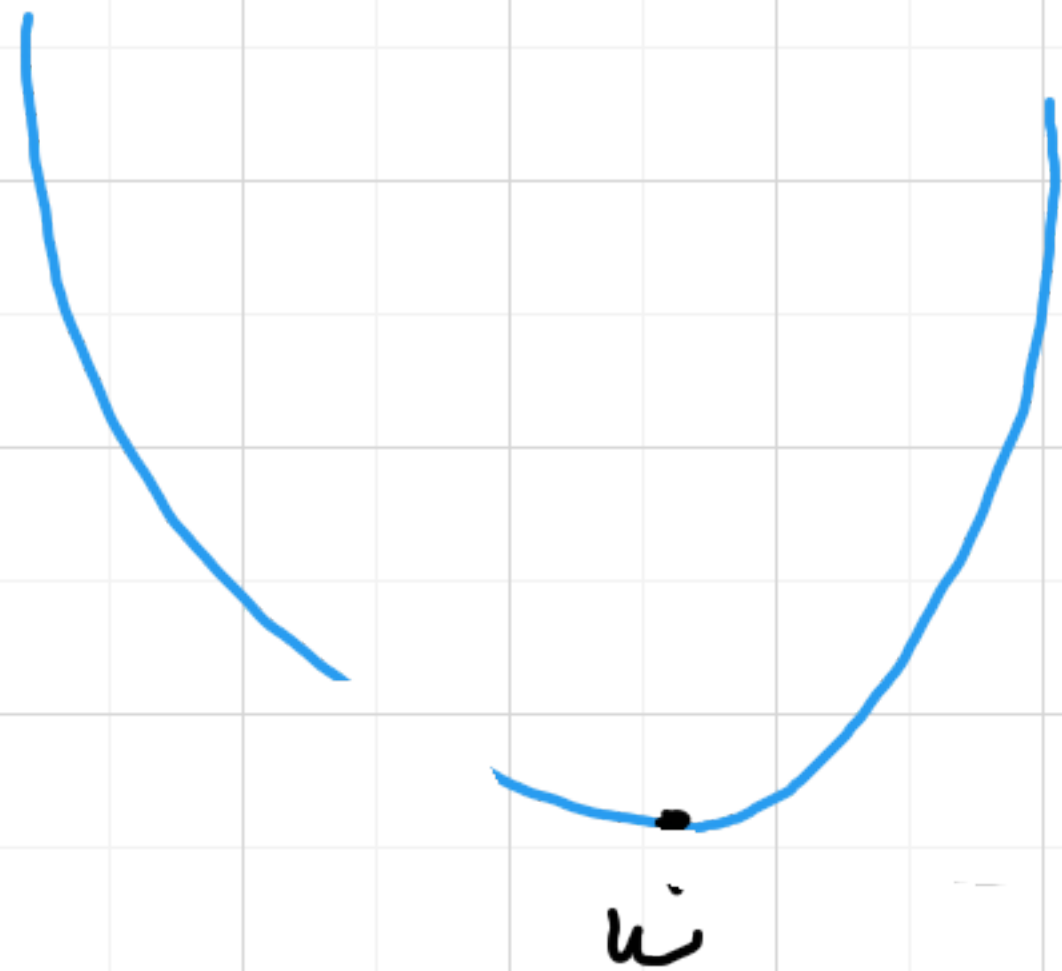
$$\frac{\partial}{\partial w} \left(\underbrace{y_i - (w x + b)}_{-x} \right)^2 =$$

$$= \frac{1}{N} \sum_i^N$$

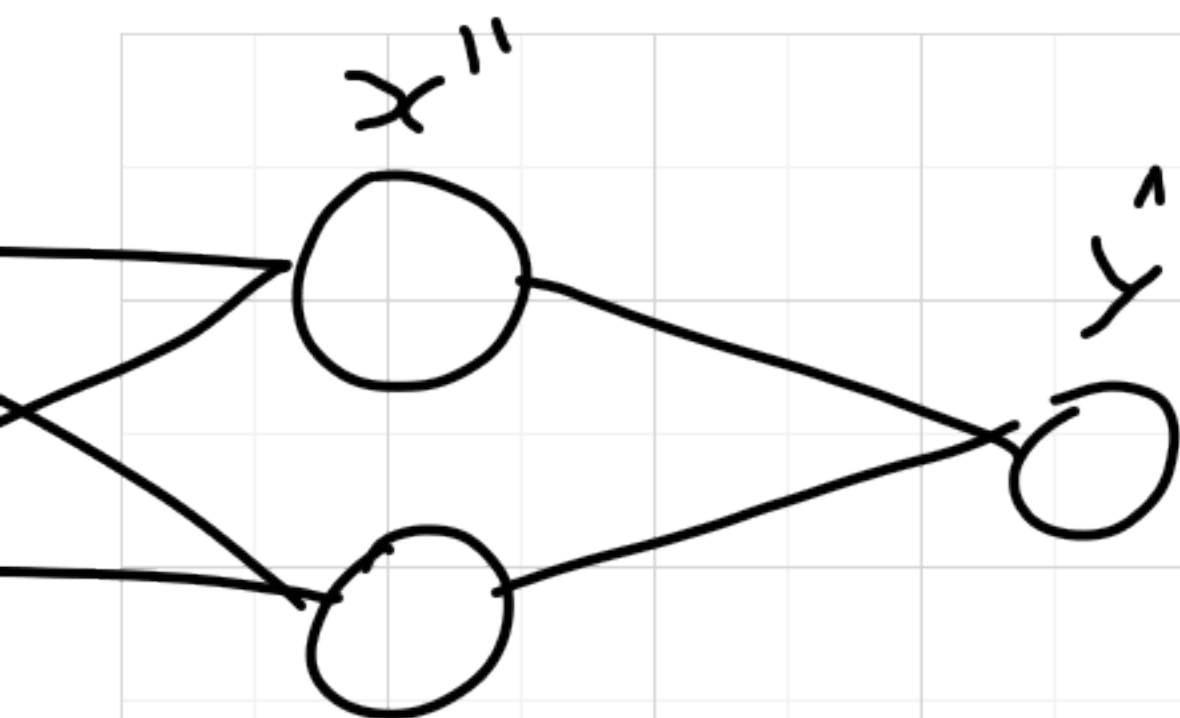
$$(-x_i)$$

error_grad = 7.5

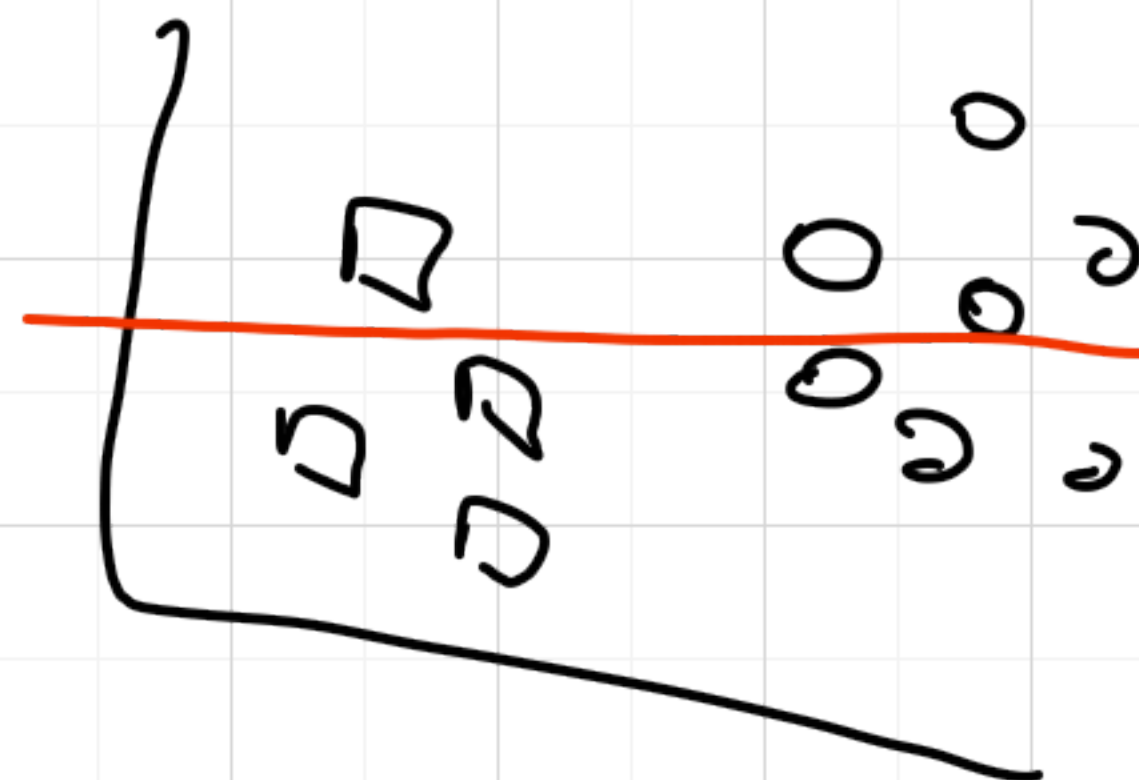
$$SF(w) \rightarrow a \cdot w^2 + b w + c$$



$$y = \frac{1}{1 + e^{-w}}$$



Compu



$$1. x'' + b$$

$$g(w_{x''} \cdot x' + b_{x''})$$

$$g(w_{x'} \cdot x + b_{x'})$$

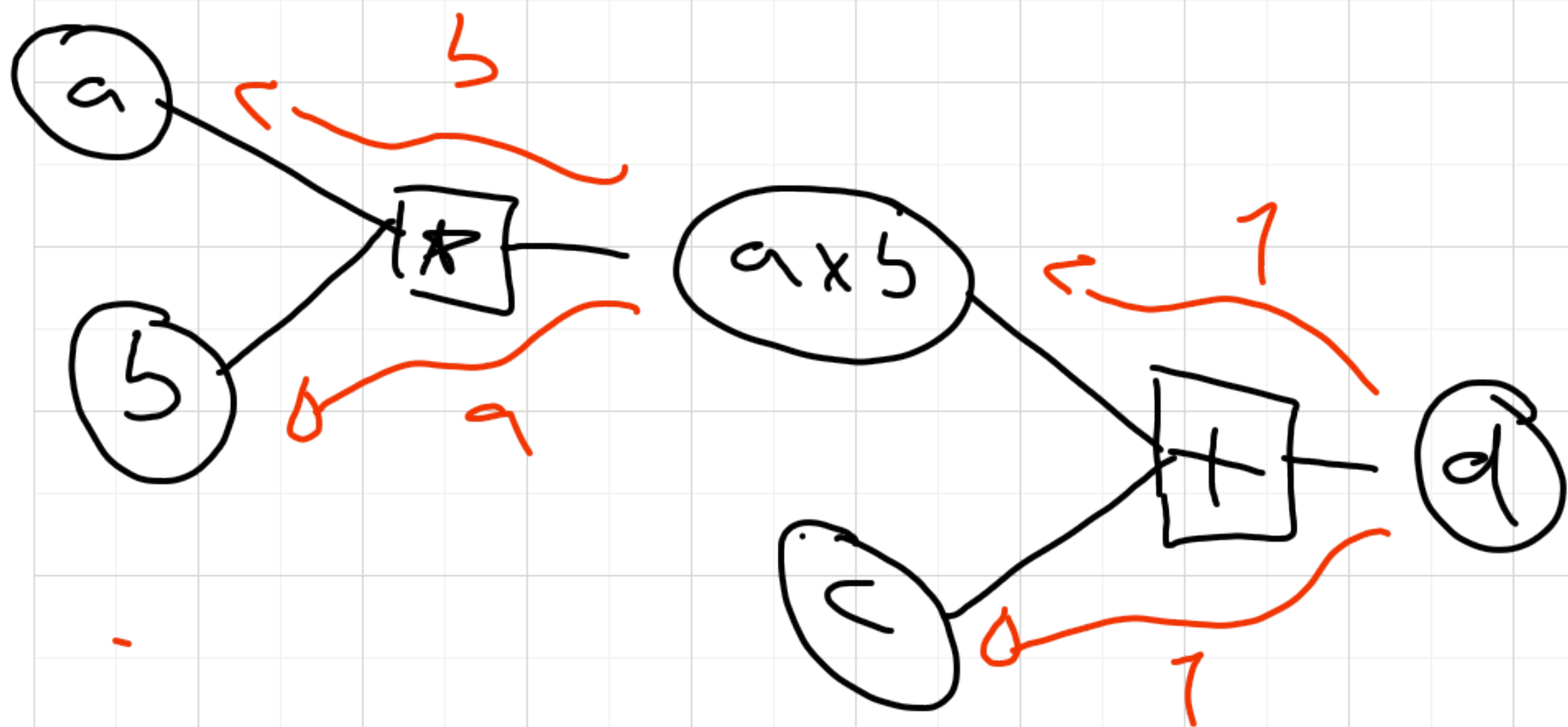
$$1 + e^{-w''} \left(\frac{1}{1 + e^{-w' \left(\frac{1}{1 + e^{-w_2}} \right)}} \right)$$

Computational Graphs

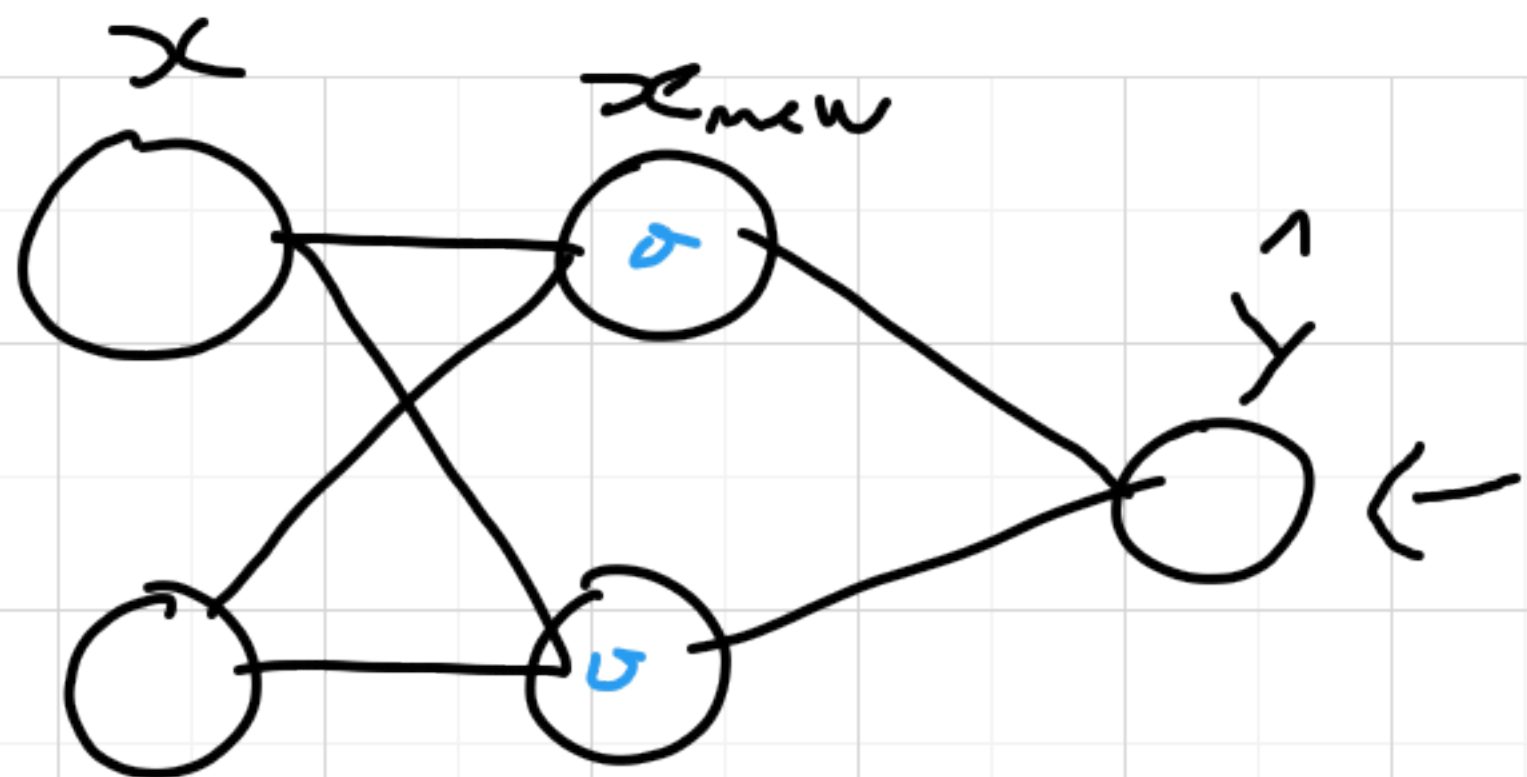
$$d = a \times b + c$$

$$d = a \times b + c$$

$$\frac{\partial d}{\partial a} = \frac{\partial d}{\partial_{\cancel{a \times b}}} \times \frac{\cancel{\partial_{a \times b}}}{\partial a} =$$



$b + c$



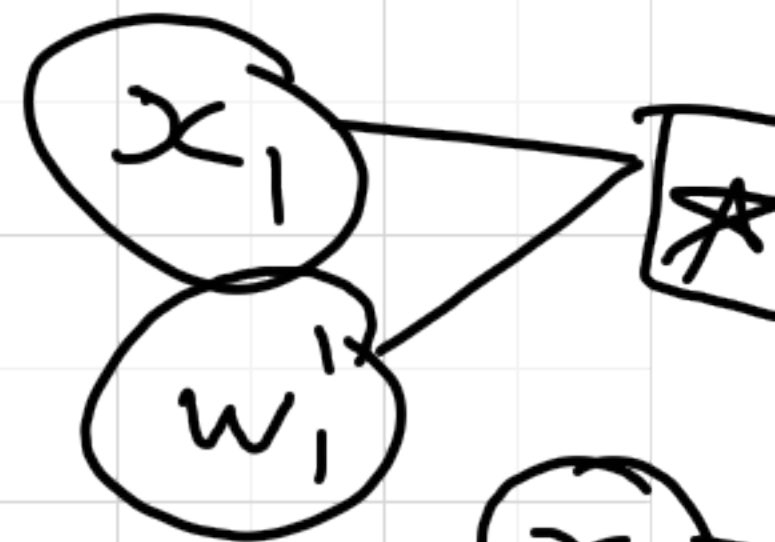
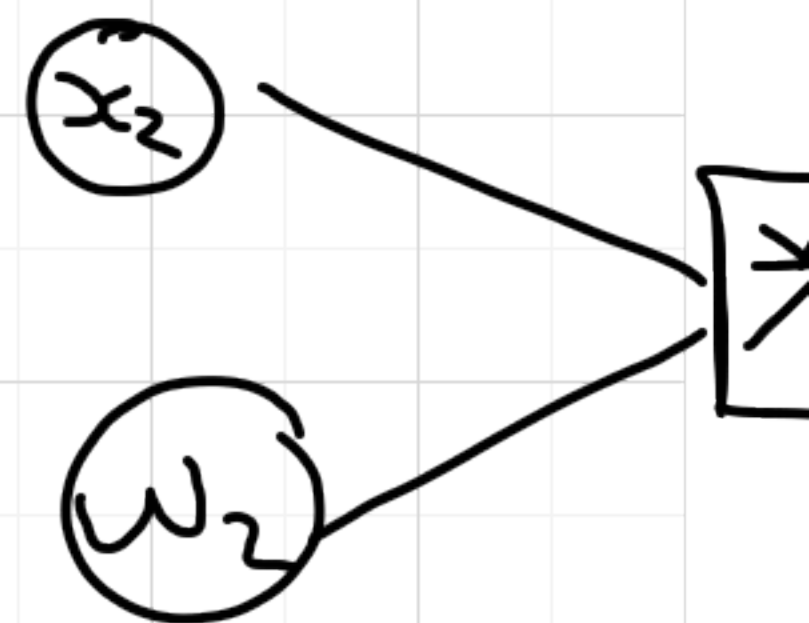
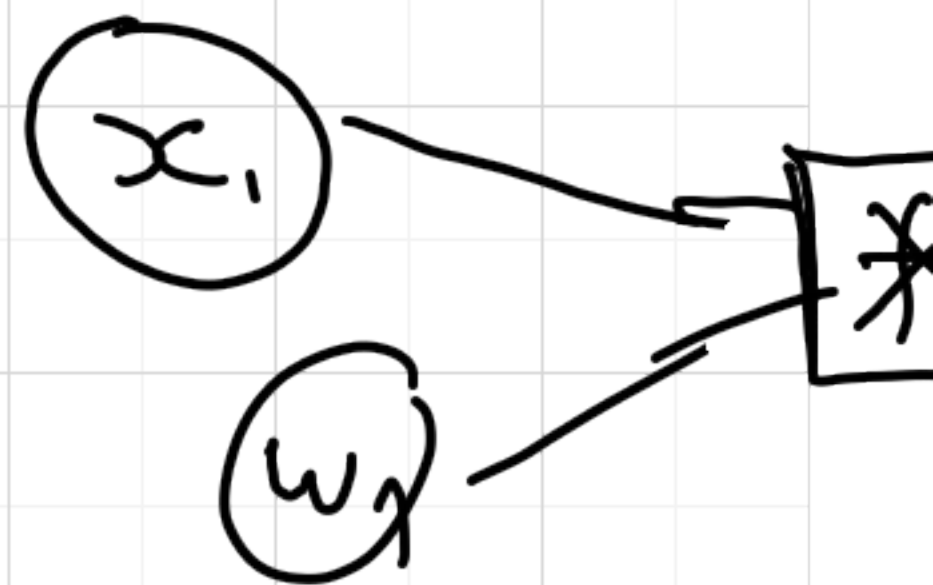
$$\sigma(x) = \text{Sigmoid}(x)$$

$$b \times 1 = b$$

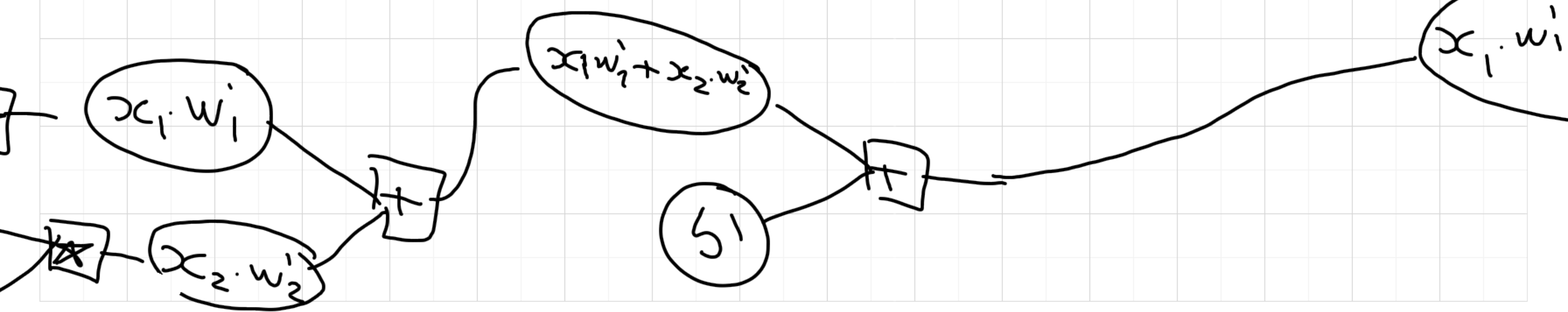
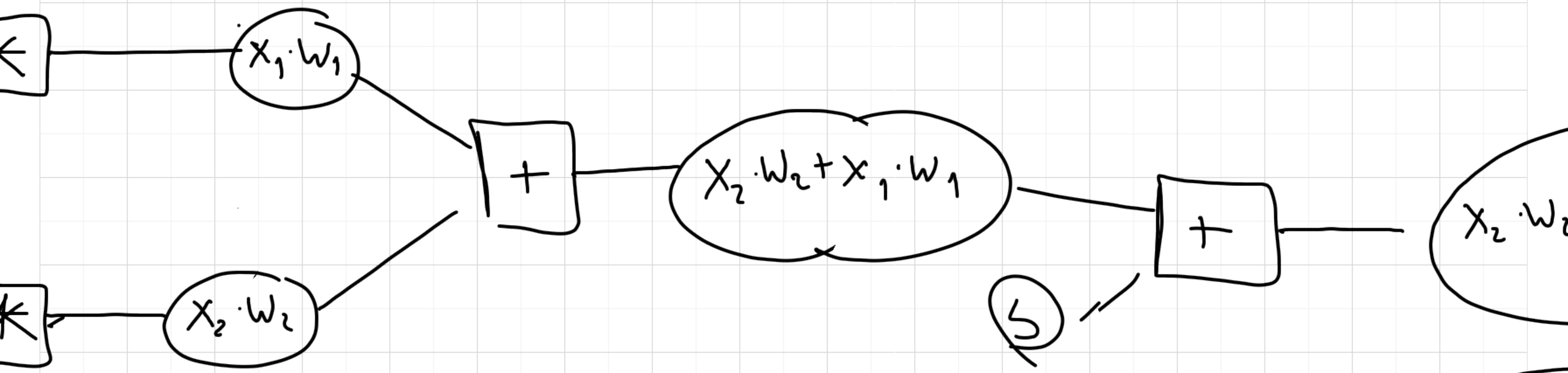
$$x_{new1} = \sigma(w_1 \cdot x_1 + w_2 \cdot x_2 + b)$$

$$x_{new2} = \sigma(w'_1 \cdot x_1 + w'_2 \cdot x_2 + b')$$

$$\hat{y} = \sigma(w''_1 \cdot x_{new1} + w''_2 \cdot x_{new2} + b'')$$

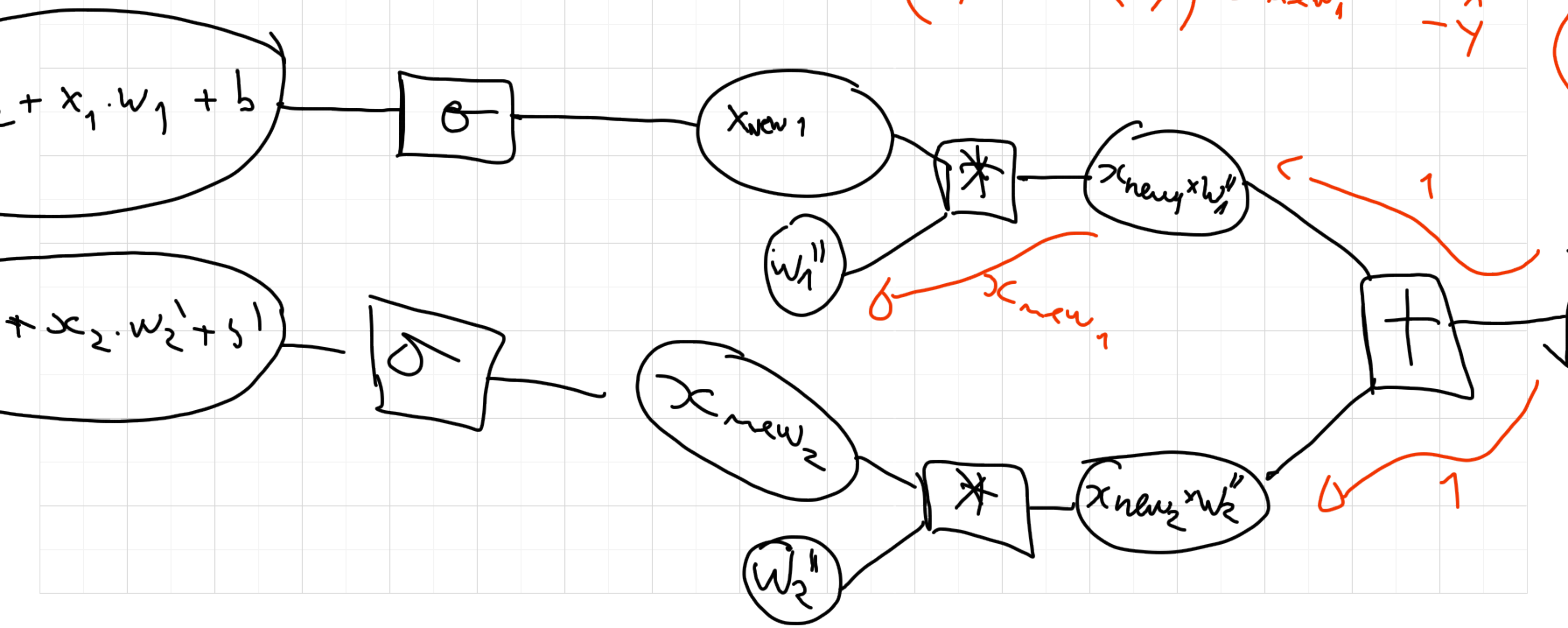


$$x_1 \times w_1 + x_2 w_2$$



$$\hat{y} = \sigma \left(w_1'' \cdot x_{new1} + w_2'' \cdot x_{new2} + b'' \right)$$

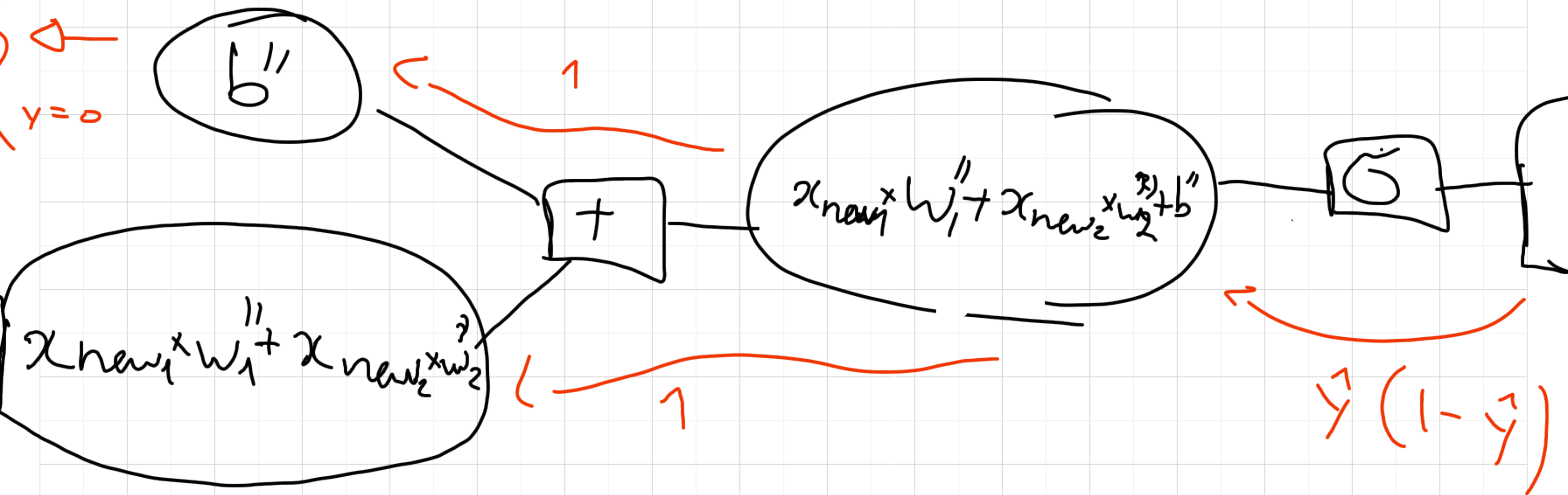
$$\frac{\partial \mathcal{L} E}{\partial w_1''} = \begin{cases} \text{if } y=1 \rightarrow (1-\hat{y}) \cdot x_{new1} & 1-\hat{y} \\ y=0 \rightarrow (-\hat{y}) \cdot x_{new1} & -\hat{y} \end{cases}$$



$$\frac{\partial CE}{\partial b''} = 1 \cdot \frac{1}{\hat{y}} \cdot (1 - \hat{y}) \cdot \begin{cases} \text{if } y=1, \frac{1}{\hat{y}} \cdot \hat{y} \cdot 1 \\ \text{if } y=0, -1 \times 1 \times \frac{1}{1-\hat{y}} \times 1 - y \end{cases} = \begin{cases} \text{if } y=1, 1 - \hat{y} \\ \text{if } y=0, -\hat{y} \end{cases}$$

$\sigma' = \sigma b)(1$

$y=1$
 $y=0$



$$x) = \frac{1}{x}$$

$$\frac{\partial C E}{\partial b} = ?$$

